

SILVER OAK UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING | EDUCATION TO INNOVATION

Course Name: Final Year Major Project (Computer Engineering)

RAG-Based Intelligent Document Chatbot Using Large Language Models

Demonstrating Document Ingestion, TF-IDF Retrieval, and Contextual Response Generation (MedRAG)

Project ID	CE-2026-RAG-03
Enrollment No	[Your Enrollment Number(s)]
Candidate Name(s)	Shivani Kuntmal / Leeza Khanjada / Parth Bavaliya
Branch / Department	B.Tech Computer Engineering
Internal Guide Name	[Prof. Guide Name] / Department Faculty

Table of Contents

01. Introduction to Research Area	07. Required Dataset, Tools & Tech Stack
02. Background and Motivation	08. Proposed System Methodology & Architecture
03. Relevance and Importance	09. Implementation & Functional Validation
04. Literature Review (Key Research)	10. Project Conclusion
05. Research Gap Analysis	11. Future Scope & Research Roadmap
06. Project Objectives	12. References (IEEE Format)

1. Introduction

Research Domain & Paradigm:	Natural Language Processing (NLP), Information Retrieval (IR), and Retrieval-Augmented Generation (RAG) using Large Language Models (LLMs).
The Document Hallucination Problem:	General-purpose LLMs generate fluent text but lack automatic access to newly uploaded proprietary documents, often producing plausible-sounding hallucinations when factual evidence is missing.
What is MedRAG?	A lightweight, locally-managed RAG-based intelligent document chatbot that ingests PDF reports, extracts text, indexes manageable chunks, retrieves evidence using TF-IDF, and grounds Google Gemini's responses.
Dual Role Interface Workflows:	Features a Patient/Student workflow for interactive natural-language document explanation and a Doctor/Admin workflow for querying and managing stored sample records.
Classroom Prototype Focus:	Prioritizes end-to-end traceability, modular 5-tier architecture, and local reproducibility without requiring complex vector database infrastructure.

2. Background and Motivation

Background of Research Work:

- **LLM Knowledge Limits:** Pre-trained LLMs have static parameter weights and cannot query newly created or private domain documents.
- **Manual Navigation Burden:** Users face immense fatigue navigating complex, multi-page PDF medical and technical reports manually.
- **Hallucination Vulnerability:** Without explicit ground truth context passed to the generation prompt, models confabulate non-factual details.
- **Infrastructure Overhead of Modern RAG:** Typical enterprise RAG stacks demand heavy GPU servers, high-cost vector databases, and complex microservices, making educational demonstration impractical.

Motivation Behind MedRAG:

- **End-to-End Traceability:** Create a completely observable engineering pipeline where each phase—ingestion, chunking, retrieval, and generation—can be monitored independently.
- **Zero-Vector Dependency:** Prove that classical sparse retrieval (TF-IDF) provides an effective, lightweight baseline for single-document QA without vector DB costs.
- **Persistent Local Storage:** Provide a robust SQLite layer to retain document chunks, metadata, and multi-turn conversational history locally.
- **Classroom Feasibility:** Develop an educational tool running on standard commodity hardware (Python/FastAPI) that clearly demonstrates foundational RAG principles.

3. Relevance and Importance

New Insights Contributed: • Decoupled 5-Tier Architecture: UI, Ingestion, Retrieval, Storage, and Generation operate as independent modules. • Explainable Evidence: TF-IDF term weights clearly expose why specific passages were chosen. • Stateful Chat: Preserves dialogue turns in local SQLite without cloud DB subscriptions.	Who is it Relevant To? • Computer Engineering Students: Provides an accessible blueprint for studying RAG mechanisms. • Healthcare Informatics Learners: Demonstrates safe report interpretation using synthetic data. • Resource-Constrained Developers: Runs on standard laptops without GPU acceleration.	Why Research is Worth Doing: • Demystifies Black-Box AI: Separates knowledge retrieval from language generation for academic auditability. • Extensible Baseline: Prepares a solid foundation for adding dense embeddings and rerankers. • Educational Safety: Explicitly bounded to synthetic data to prevent unauthorized clinical claims.
--	--	--

4. Literature Review

1	Lewis et al. [1] (NeurIPS 2020)	Retrieval-Augmented Generation (RAG)	Natural Questions, TriviaQA	SOTA open-domain QA	Parametric generator + non-parametric retrieval memory. Conceptual foundation for our project.
2	Guu et al. [2] (ICML 2020)	REALM: Latent Knowledge Retrieval	Open-domain QA corpora	High zero-shot QA accuracy	Separated knowledge retrieval from generation. Informs MedRAG's modular architecture.
3	Karpukhin [3] (EMNLP 2020)	Dense Passage Retrieval (DPR)	Wikipedia 21M benchmark	Top-20: 78.8% vs BM25 59.1%	Contrasted dense vs sparse retrieval. Motivates MedRAG's future dense embedding migration.
4	Yan et al. [8] (CRAG 2024)	Corrective RAG (CRAG)	Multi-hop QA benchmarks	Factual precision gains	Retrieval evaluation & correction before generation. Inspires future handling of poor query matches.

5. Research Gap Analysis

Gap 1: Prohibitive Infrastructure Overhead	Commercial RAG pipelines depend on GPU clusters and hosted vector databases, creating high barrier to entry for classroom and lightweight local deployments.
Gap 2: Lack of Observable End-to-End Systems	Most research isolates mathematical sub-components on benchmarks rather than providing an end-to-end engineering pipeline from raw PDF to interactive dialogue.
Gap 3: Lexical vs. Semantic Paraphrase Vulnerability	Classical TF-IDF suffers when user queries use synonyms absent in the source text. An explainable baseline is required before deploying black-box embeddings.
Gap 4: Unregulated Medical QA Tool Deployments	Many tools claim medical decision support without boundaries. An educational demonstrator must explicitly use synthetic reports and disclaim clinical advice.

6. Project Objectives

Primary Goal:	To construct and validate a lightweight, locally-executable RAG-based document chatbot prototype (MedRAG) demonstrating PDF ingestion, TF-IDF text retrieval, and LLM-grounded explanation generation.
Objective 1: Ingestion & Chunking	Implement embedded text extraction from uploaded PDF reports and partition content into focused chunks while tracking source metadata.
Objective 2: Sparse Lexical Retrieval	Develop an explainable TF-IDF retrieval system that transforms natural-language queries into term-frequency vectors and ranks document chunks.
Objective 3: Persistent Local Storage	Design an SQLite schema (medrag.db) to persistently store uploaded reports, chunk indices, and multi-turn conversational dialogue.
Objective 4: Contextual LLM Prompting	Integrate Google Gemini LLM via server-side API handlers to formulate grounded answers strictly based on retrieved document excerpts, minimizing hallucinations.

7. Required Dataset, Tools & Technology

Python 3.10–3.12	Core application programming language
FastAPI	High-performance web API layer & endpoint routing
Uvicorn	ASGI server for local development execution (http://127.0.0.1:8000)
PDF Text Extractor	Extracts embedded text layers from uploaded PDF files
TF-IDF Vectorizer	Sparse term-weighted lexical retrieval of document chunks
Google Gemini API	Large Language Model for contextual natural-language generation
SQLite (medrag.db)	Local persistent database for reports, chunks, & chat history
Python unittest	Automated functional testing and component verification

Dataset & Demonstration Scope:

- **Synthetic Medical Reports:** Bundled sample report for demonstration patient 'Aarav Sharma' (Identifier: DEMO-003).
- **Strict Educational Boundaries:** Does not use real patient PHI. Explicitly claims NO clinical diagnosis, treatment advice, or FDA compliance.
- **Secure API Credentialing:** Gemini API key kept securely in server-side .env file; never exposed to client browser.
- **Repeatable Demonstrations:** Resetting database re-indexes synthetic sample instantly.

8. System Methodology & Architecture



Layer 1 — Ingestion & Extraction:	PDF reports with embedded text layers are parsed; text is sanitized and associated with persistent report metadata.
Layer 2 — Chunking & Indexing:	Long documents are subdivided into smaller searchable segments to avoid prompt context overflow and preserve provenance.
Layer 3 — Sparse TF-IDF Ranking:	Documents and queries are transformed into term-frequency inverse-document-frequency vectors; cosine similarity selects top-k chunks.
Layer 4 — Augmentation & Generation:	Retrieved excerpts and dialogue history are formatted into a constrained prompt, instructing Google Gemini to respond strictly from evidence.
Layer 5 — Local Data Persistence:	SQLite tracks uploaded documents, generated chunks, and conversation dialogue for complete multi-turn session recall.

9. Results & Functional Validation

Synthetic PDF Processing	Extract and index embedded text; verified on bundled sample report	PASSED
Relevant Evidence Retrieval	Return relevant chunks for query; TF-IDF retrieval verified	PASSED
Grounded Response Generation	Generate answer using retrieved evidence + Gemini LLM	PASSED
Report Search by ID / Name	Locate sample reports by patient name or DEMO-003	PASSED
Conversation Turn Persistence	Stored and restored locally across sessions in medrag.db	PASSED
Image-Only Scanned PDF	Not supported in v1 (No OCR layer; documented boundary)	Bounded

Academic Rigor in Evaluation: The paper consciously distinguishes functional validation from benchmark evaluation. Because a controlled, labeled question test set was not constructed, no invented accuracy percentages are reported. The prototype successfully proves that a lightweight, decoupled RAG pipeline can execute reliably on commodity hardware.

10. Conclusion

Successful Prototype Realization	Engineered MedRAG, a fully functional, reproducible RAG-based document chatbot combining PDF parsing, TF-IDF retrieval, SQLite persistence, and Google Gemini response generation.
Accessible Engineering Baseline	Demonstrated that an effective, transparent document QA system can be executed locally without specialized vector infrastructure or complex embedding pipelines.
Academic Honesty & Transparency	Distinctly differentiated between functional engineering verification and benchmark performance, setting clear boundaries regarding synthetic medical demonstration.
Solid Foundation for Scalable Evolution	Provided an extensible 5-tier architecture that allows future researchers to seamlessly swap sparse TF-IDF with dense semantic embeddings and corrective retrieval.

11. Future Scope & Research Roadmap

1. Dense Semantic Embeddings:	Transition from sparse TF-IDF to dense vector representations (Chroma / FAISS / Sentence-Transformers) to capture semantic relationships.
2. Hybrid Retrieval & Re-ranking:	Combine lexical TF-IDF with dense semantic search, followed by a cross-encoder re-ranker for optimal chunk prioritization.
3. Corrective RAG (CRAG) & Self-RAG:	Implement mechanisms inspired by CRAG [8] and Self-RAG [7] to evaluate retrieved passage quality before triggering generation.
4. OCR for Scanned Documents:	Integrate Tesseract OCR or PaddleOCR to support scanned medical documents and physical lab sheets.
5. Benchmark Evaluation on PubMedQA:	Conduct controlled quantitative benchmarking against PubMedQA [9] to measure Recall@k, answer correctness, and latency.

12. References (IEEE Format)

- [1] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.
- [2] K. Guu et al., "Retrieval augmented language model pre-training," in Proc. 37th Int. Conf. Mach. Learn. (ICML), vol. 119, pp. 3929–3938, 2020.
- [3] V. Karpukhin et al., "Dense passage retrieval for open-domain question answering," in Proc. 2020 Conf. EMNLP, pp. 6769–6781, 2020.
- [4] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," in Proc. 16th Conf. EACL, pp. 874–880, 2021.
- [5] S. Borgeaud et al., "Improving language models by retrieving from trillions of tokens," in Proc. 39th Int. Conf. Mach. Learn. (ICML), vol. 162, pp. 2206–2240, 2022.
- [6] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.
- [7] A. Asai et al., "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in Proc. ICLR, 2024.
- [8] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," arXiv:2401.15884, 2024.
- [9] Q. Jin et al., "PubMedQA: A dataset for biomedical research question answering," in Proc. EMNLP-IJCNLP, 2019.
- [10] E. Papagiannopoulou et al., "Large-scale semantic indexing and question answering in biomedicine," in Proc. 2016 IEEE BIBM, 2016.